

Problem Set 8

C. Magic Gems

tianzheyu submissions

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#	When	Who	Problem	Lang	Verdict	Time	Memory
385713700	Aug/05/2026 14:18UTC+10	tianzheyu	D - Two Divisors	C++23 (GCC 14-64, msys2)	Accepted	1000 ms	39400 KB

Process

I used the Sieve of Eratosthenes to precompute the Smallest Prime Factor for fast queries. For each number, I extracted the highest power of its smallest prime factor as d_1 , and the remaining part as d_2 . This guarantees that d_1 and d_2 share no common prime factors, satisfying $\gcd(d_1 + d_2, a_i) = 1$.

Challenges and Overcoming

I thought my code was bugged because my terminal output for 24 was 8 3, while the official example showed 2 3. I found this when directly comparing my results with the problem statement. I soon realized my answer was perfectly valid ($\gcd(8+3, 24) = 1$). The bug was simply me forgetting that this is a Special Judge problem where multiple outputs are correct. Next time to prevent this, I will carefully read the problem notes and manually verify the mathematical properties of my output rather than just string matching the example.